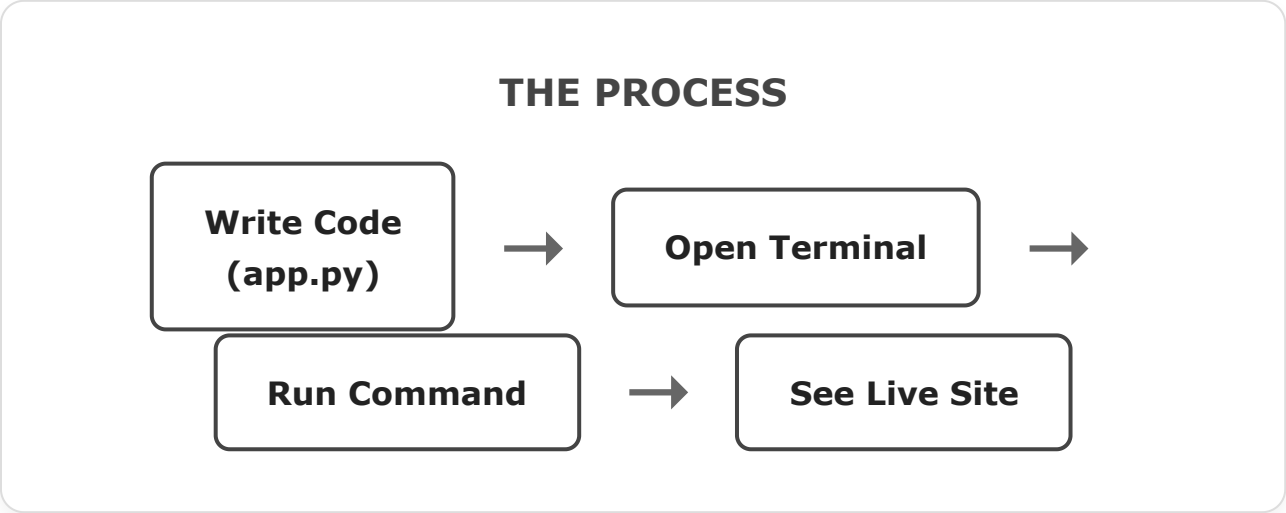
2. STREAMLIT BASICS

THE PLAYGROUND

Before we build the big Loan App, we need to learn the tools. We are using a library called **Streamlit**. It lets you turn Python scripts into websites in minutes!

In this worksheet, we will learn the "bricks" you can use to build your website: Buttons, Text, Checkboxes, and Layouts.

PHASE 1: YOUR FIRST APP



> 💡 **SIMPLE EXPLANATION:**
>
> *To make a Streamlit app, you just write a normal Python file. The magic happens when you **run** it from the terminal.*

STEP 1: CREATE A FILE NAMED APP.PY

PYTHON CODE

```python
import streamlit as st

st.title("Hello World!")
```

STEP 2: RUN IT!

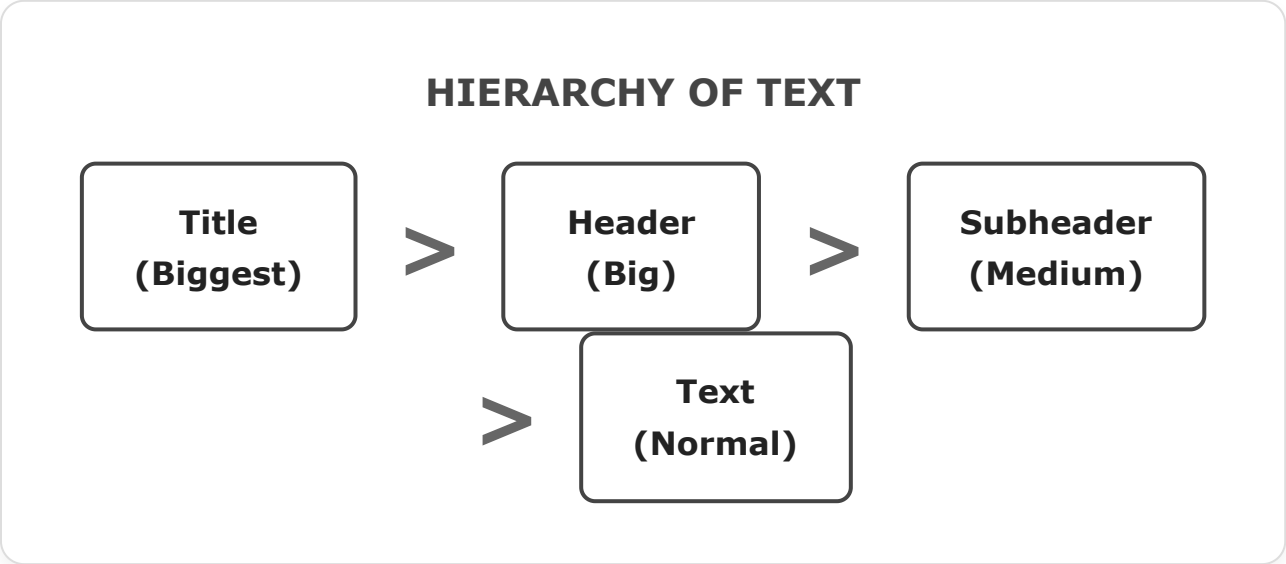
Open your terminal (command prompt), go to the folder where you saved the file, and type:

PYTHON CODE

```python
streamlit run app.py
```

A new tab will open in your browser showing your app!

PHASE 2: WRITING TEXT



> 💡 **SIMPLE EXPLANATION:**
>
> *Streamlit has different commands for different sizes of text. Always start with `st.title` for your main heading.*

TRY THIS CODE:

PYTHON CODE

```python
import streamlit as st

st.title("Streamlit Demo by Bipul")

st.header("Header for the App")

st.subheader("Sub-Header of the app")

st.text("This is a example Text")
```

PHASE 3: SUCCESS & ERROR ALERTS

FEEDBACK TYPES

Success (Green)	Warning (Yellow)	Error (Red)	Info (Blue)

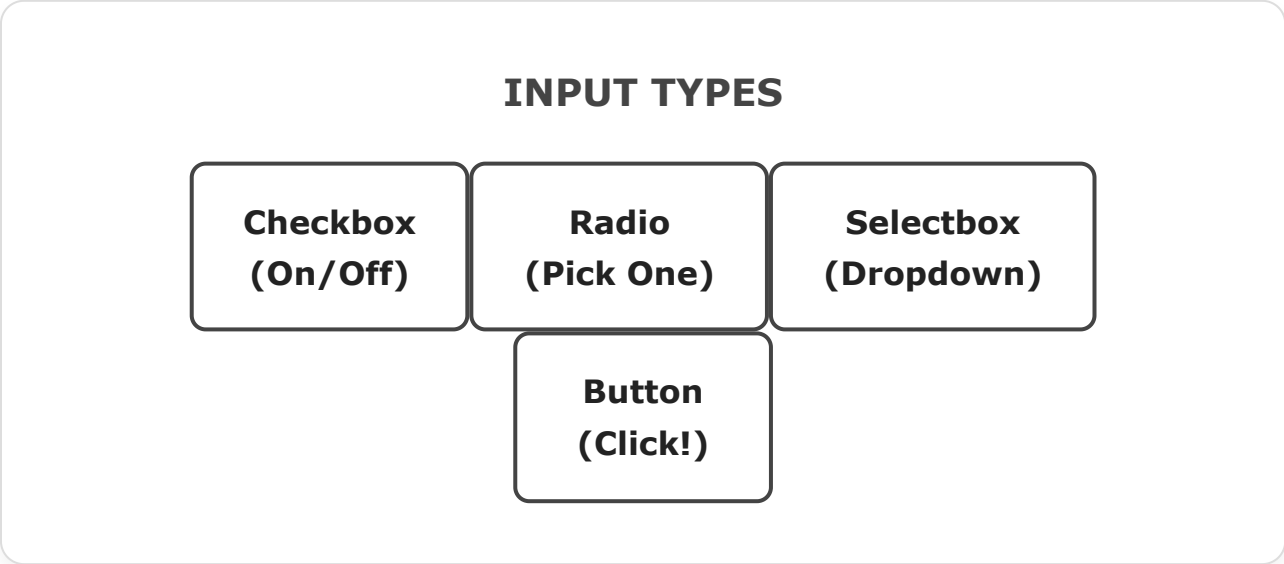
> 💡 **SIMPLE EXPLANATION:**
>
> *Websites need to talk to users. Did their submission work? Did it fail? Use these colorful boxes to give feedback.*

TRY THIS CODE:

PYTHON CODE

```python
st.success("Success")
st.warning("Warning")
st.info("Information")
st.error("Error")
```

PHASE 4: INTERACTIVE WIDGETS



> 💡 **SIMPLE EXPLANATION:**
>
> ***Widgets*** *allow users to interact with your app.*
> - ***Checkbox****: True or False.*
> - ***Radio/Selectbox****: Choosing from a list.*
> - ***Button****: Triggers an action when clicked.*

1. THE CHECKBOX

PYTHON CODE

```python
# Checkbox returns True if checked
if st.checkbox("Select/Unselect"):
    st.text("User selected the checkbox")
else:
    st.text("User has not selected the checkbox")
```

2. RADIO BUTTONS

PYTHON CODE

```python
# Radio returns the text of the selected option
state = st.radio("What is your favorite color?", ("Red", "Green", "Blue"))

if state == 'Green':
    st.success("That is my favorite color as well")
```

3. SELECT BOX (DROPDOWN)

PYTHON CODE

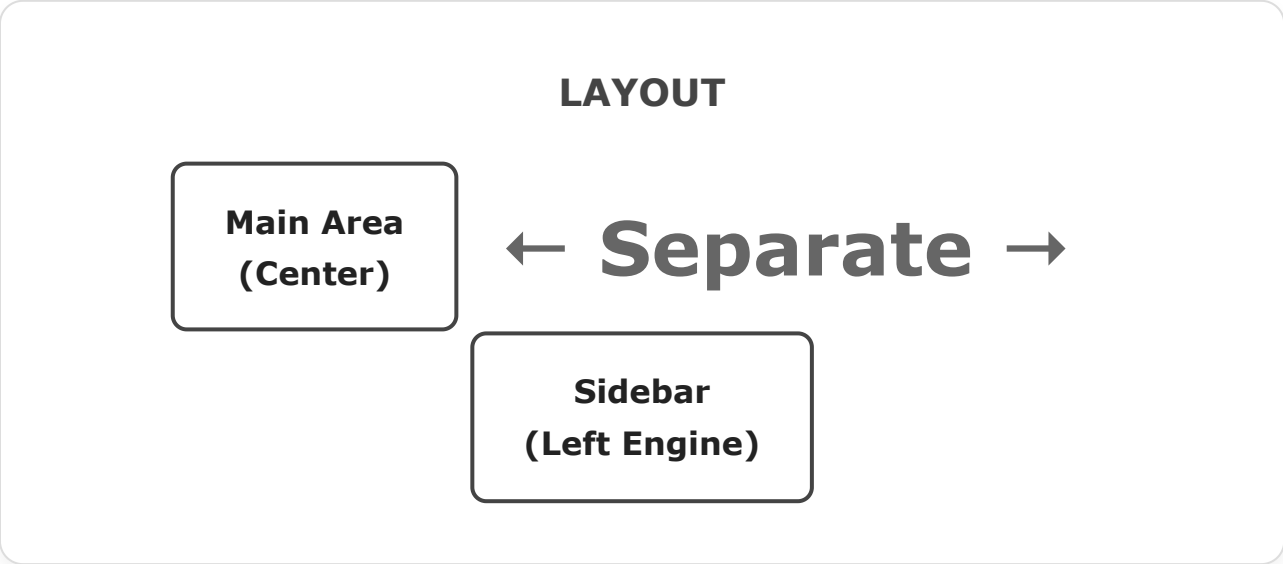
```python
occupation = st.selectbox("What do you do?", ["Student", "Vlogger", "Engineer"])

st.text(f"Selected option is {occupation}")
```

4. THE BUTTON

PYTHON CODE

```python
if st.button("Example Button"):
    st.success("You clicked it")
```

PHASE 5: THE SIDEBAR



> 💡 **SIMPLE EXPLANATION:**
>
> *The **Sidebar** is a special area on the left. It's great for controls, filters, or navigation, keeping the main area clean. Just add `.sidebar` to any command!*

TRY THIS CODE:

PYTHON CODE

```python
# Add things to the sidebar using st.sidebar
st.sidebar.header("Heading of Sidebar")
st.sidebar.text("MLOPs by FunctionUP")
```